

DBMS — Transaction States

Core Database Management Concepts & Exam Revision Guide

QUICK REFERENCE

A **transaction state** represents the execution phase through which a database transaction moves during its lifecycle—from execution to commit or rollback.

1. State Breakdown

1. ACTIVE ●

Definition: Entered as soon as the transaction begins execution.

Operations: Performs READ, WRITE, and CPU arithmetic inside RAM/volatile memory.

```
READ(A)
A = A - 50
WRITE(A)
```

RULE

Start → ACTIVE

2. PARTIALLY COMMITTED ●

Definition: Reached when $N - 1$ operations are complete, leaving only final COMMIT pending.

Location: Data changes reside in RAM/cache memory and are not yet on permanent disk storage.

```
WRITE(B)
-- [Partially Committed]
COMMIT ← Final step
```

GOLDEN RULE

All ops done except COMMIT

3. COMMITTED ●

Definition: Triggered when COMMIT successfully completes.

Impact: Updates are flushed to permanent non-volatile storage (Hard Disk/SSD). Changes are permanent.

EXAM TRAP

Never transition from COMMITTED to FAILED!

4. TERMINATED ●

Definition: Final state after completion or complete rollback.

Deallocation: Releases CPU, RAM, registers, locks, and network bandwidth back to the OS.

KEY CONCEPT

Termination = Resource Release

2. Failure & Abort Path

5. FAILED ●

Triggers: Hardware/software crashes, network loss, system outage, deadlock, or assertion checks.

Transitions: Can occur directly from **ACTIVE** or **PARTIALLY COMMITTED** (before commit flushes to disk).

6. ABORT & ROLLBACK ✗

Action: Reverts the database to its previous consistent state (Rollback).

Restart Rule: A failed transaction is never resumed from the crash point; it is restarted as a *new execution*.

3. State Transition Flowchart



4. Comprehensive State Summary

State	What Happens Under the Hood?	Key Exam Keyword
Active	Transaction is executing operations (READ/WRITE) in CPU & RAM.	EXECUTION
Partially Committed	All operations are completed except the final COMMIT. Changes are still buffered in volatile memory.	ALMOST DONE
Committed	COMMIT succeeds. Intermediate state is permanently written to non-volatile storage (disk).	PERMANENT
Terminated	Transaction exits system execution. Memory buffers, locks, and system resources are freed.	DEALLOCATION
Failed	An error or crash prevents normal continuation during active or partially committed states.	FAILURE
Aborted	Transaction is cancelled, changes rolled back, and system prepares for a clean restart.	ROLLBACK

5. High-Yield Exam Traps & Pitfalls

 **Trap 1: N Operations Completed = Partially Committed?**

Correct Understanding: Only if COMMIT is explicitly the *N-th* operation. The rule is strictly **N - 1 operations completed** with only the COMMIT remaining.

 **Trap 2: Can a Failed transaction directly become Active again?**

Correct Understanding: NO! The trajectory MUST follow: FAILED → ABORT → ROLLBACK → RESTART. A transaction never resumes from the precise point of failure.

 **Trap 3: Transitioning after COMMITTED?**

Correct Understanding: Once committed, a transaction can **ONLY** transition to TERMINATED. You cannot draw an arrow from COMMITTED to FAILED or ABORTED.

 **Trap 4: Where can Failure occur?**

Correct Understanding: Failure occurs either while in **ACTIVE** state or **PARTIALLY COMMITTED** state (if power fails while flushing changes to disk).

6. Super-Fast Memory Mnemonics

 **Rapid Recall Formulas**

Standard Success Path:

A → P → C → T

Active → Partially Committed → Committed → Terminated

Failure Recovery Path:

F → A → R

Failed → Abort → Rollback / Restart

 **15-SECOND EXAM SUMMARY**

Start → Active (Executing) → Almost Done (Partially Committed) → Commit (Committed to Disk) → Release (Terminated)